

Rational Hodge classes on mixed products of K3 moduli spaces

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Abstract

Combining OpenAI's rational Hodge theorem for arbitrary mixed products of projective K3 surfaces with Bülles' Chow-motive decomposition, we record rational Hodge in every codimension for finite mixed products of the smooth projective K3 moduli spaces covered by that decomposition. The factors may have different K3 bases and include Hilbert schemes, twisted stable-sheaf moduli, and eligible moduli of Bridgeland-stable objects. We give the explicit pull-push argument, retaining smoothness, projectivity, and stability hypotheses. The contribution is an assembled corollary of the cited results; it supplies neither a new proof of the base K3 theorem nor a new motivic reduction.

1 Statement and provenance

For a smooth projective complex variety X , write

$$\mathrm{Hdg}^p(X) = H^{2p}(X, \mathbb{Q}) \cap H^{p,p}(X, \mathbb{C}).$$

Rational Hodge in codimension p means surjectivity of $\mathrm{cl} : \mathrm{CH}^p(X)_{\mathbb{Q}} \rightarrow \mathrm{Hdg}^p(X)$. Here Chow groups and all correspondences have rational coefficients.

Theorem 1. *Let $S_1, \dots, S_r$ be complex projective K3 surfaces and let $\alpha_i \in \mathrm{Br}(S_i)$. For each i , let M_i be a smooth projective moduli space of Gieseker-stable α_i -twisted sheaves, or a smooth projective moduli space of σ_i -stable objects in $D^b(S_i, \alpha_i)$ for a generic Bridgeland stability condition σ_i , as in Bülles' Theorem 0.1 [3]. Then the rational Hodge conjecture holds in every codimension on*

$$X = \prod_{i=1}^r M_i.$$

The K3 bases may be different or repeated. In particular, the conclusion holds for every finite mixed product of Hilbert schemes $S_i^{[n_i]}$, $n_i \geq 0$.

The sheaf data and polarization defining each Gieseker moduli problem are fixed. Genericity for Bridgeland stability is relative to the fixed numerical data of that moduli problem. Smoothness and projectivity of the actual stable moduli space are hypotheses, rather than conclusions about arbitrary numerical data or stability conditions. The theorem does not add a primitivity or fine-moduli assumption to Bülles' statement. Its exact stable-object scope is that of the cited theorem.

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The principal input is OpenAI’s theorem [7, Theorem 1.1] that rational Hodge holds on every finite product of projective complex K3 surfaces, with arbitrary distinct or repeated factors. We use the public OpenAI Math Release snapshot at the following commit, accessed October 6, 2026:

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

Manuscript cover dates in the references are not assertions about their earliest public disclosure.

Arapura had already proved conditional Hodge transfers to Hilbert schemes and K3 sheaf moduli [1, Theorems 5.4 and 5.7(3)], and a broader transfer for motivated varieties [2, Lemma 4.2 and Theorems 7.4, 7.8]. Bülles supplies the precise unconditional Chow-motive splitting for twisted sheaves and the eligible Bridgeland moduli [3, Theorem 0.1]. The earlier OpenAI quadratic-locus companion already records the moduli and self-power application on its restricted locus [5, Corollary 9.5]. Its general exact Kuga–Satake criterion, together with the later universal Kuga–Satake theorem [6], also yields the unrestricted-base self-power application. We record the mixed-moduli assembly: the extra mixed-base quantifier is supplied by [7], and the reduction itself is inherited. We make no claim of a first proof or substantial independent novelty.

2 The transfer with degrees made explicit

For a smooth projective V of dimension d_V and a smooth projective W , a correspondence $A \in \mathrm{CH}^a(V \times W)_{\mathbb{Q}}$ acts by

$$A_* z = (\mathrm{pr}_W)_*(\mathrm{pr}_V^* z \cdot A).$$

On cohomology this sends a rational (p, p) class to a rational $(p + a - d_V, p + a - d_V)$ class. Cycle classes commute with this action. If $B \in \mathrm{CH}^b(W \times V)_{\mathbb{Q}}$, then $B \circ A$ has codimension $a + b - \dim W$ in $V \times V$.

Proposition 2. *Suppose a smooth projective X is a direct summand of a finite sum of Tate-twisted rational Chow motives of smooth projective varieties K_j . If rational Hodge holds in every codimension on each K_j , it holds in every codimension on X .*

Proof. The split maps and projectors give homogeneous rational algebraic correspondences

$$A_j \in \mathrm{CH}^{a_j}(X \times K_j)_{\mathbb{Q}}, \quad B_j \in \mathrm{CH}^{b_j}(K_j \times X)_{\mathbb{Q}},$$

with

$$a_j + b_j = \dim X + \dim K_j, \quad \sum_j B_j \circ A_j = \Delta_X. \quad (1)$$

Using the actual cycle codimensions avoids any convention-dependent sign for a Tate twist. Let $z \in \mathrm{Hdg}^p(X)$ and put $q_j = p + a_j - \dim X$. The class $A_{j*} z$ lies in $\mathrm{Hdg}^{q_j}(K_j)$. If q_j is in the cohomological range, choose $Z_j \in \mathrm{CH}^{q_j}(K_j)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_j) = A_{j*} z$ by the assumed Hodge conjecture. Outside that range the target is zero, and take $Z_j = 0$. Then

$$Z = \sum_j B_{j*} Z_j \in \mathrm{CH}^p(X)_{\mathbb{Q}}, \quad \mathrm{cl}(Z) = \sum_j B_{j*} A_{j*} z = z.$$

The codimension is $q_j + b_j - \dim K_j = p$ by (1), and the final equality uses the diagonal identity. $\square$

Proof of Theorem 1. First assume that all factors are nonempty and positive dimensional. Bülles gives split maps

$$h(M_i) \longrightarrow \bigoplus_{j \in J_i} h(S_i^{k_{ij}})(t_{ij}) \longrightarrow h(M_i)$$

whose composite is the identity, with finite J_i , $1 \leq k_{ij} \leq \dim M_i$, and integers t_{ij} . Tensor these maps in rational Chow motives and distribute the finite sums. The motive of X is a summand of a finite sum of Tate twists of motives of

$$K_J = \prod_{i=1}^r S_i^{k_{ij}}, \quad J = (j_1, \dots, j_r) \in \prod_i J_i.$$

At the level of correspondences these are exterior products of the individual split maps; their composite sum is the exterior product of the diagonals, hence Δ_X . Their codimensions add, so they satisfy (1). The rational Hodge theorem for mixed K3 products [7] applies to every K_J . Proposition 2 therefore proves the claim.

An empty factor gives an empty product. A smooth projective zero-dimensional factor over $\mathbb{C}$ is a finite disjoint union of points; remove it on each component and apply the preceding argument. The empty product is a point. Codimensions outside $[0, \dim X]$ have zero target. For $n \geq 1$, $S^{[n]}$ is the smooth projective moduli of rank-one ideal sheaves with Mukai vector $(1, 0, 1 - n)$ and is in the untwisted sheaf case. For $n = 0$ it is a point. This proves the stated Hilbert-scheme consequence. $\square$

Remark 3 (Non-fine and twisted moduli). *The split maps used here are Bülles’ algebraic Chow correspondences. We do not construct them by presuming that an honest universal family exists. The cited construction uses quasi-universal/twisted data and the normalized diagonal identity underlying Markman’s result [4, Section 3]. Raw duplication of a universal family does not preserve its Ext diagonal formula: multiplicities must be normalized. The present argument needs only the actual rational motive splitting, which already covers the Brauer classes in Theorem 1.*

3 Scope and verification

The transfer explicitly uses Hodge for mixed products K_J ; separate self-power theorems alone do not supply this premise. Conversely, $S^{[1]} = S$ shows that the unrestricted mixed-moduli conclusion includes the mixed-K3 base theorem. Thus, given Bülles’ established reduction, these two universal assertions are equivalent. This explains exactly where the principal mathematical input lies.

No claim is made about arbitrary deformations of $K3^{[n]}$ type, singular moduli or semistable compactifications, moduli on arbitrary abelian surfaces, the integral or generalized Hodge conjecture, or finite-dimensionality of Chow motives.

The verification materials distinguish the explicit correspondence argument from the unformalized geometric input in [7]. The upstream Kuga–Satake Lean module checks rational Clifford/tensor identities; it does not formalize algebraic cycles, the universal Kuga–Satake theorem, or the theorem of this note. Reproducible small linear-algebra and sign checks are supporting evidence for parts of the upstream audit, not certificates for its full geometry.

AI use and review status. AI tools were used extensively in research, source and priority checks, drafting, and verification. The cited OpenAI manuscripts are attributed to OpenAI according to their supplied citation information. This note has not undergone conventional human peer review or refereeing. Automated adversarial reviews are not human peer review.

References

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